

TrES-2b

R-1140

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_210526 · created 2026-07-31T00:02:01+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459360.86597 +/- 0.00079 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.043
Transit depth (Rp/Rs)^2	2.22 +/- 1.28 [%]
Orbital Inclination (inc)	86.79 +/- 2.85
Airmass coefficient 1 (a1)	1.216 +/- 0.1
Airmass coefficient 2 (a2)	-0.17 +/- 0.24
Scatter in the residuals of the lightcurve fit is	8.3 %
Best Comparison Star	#3 - [455, 77]
Optimal Aperture	2.0
Optimal Annulus	10.38
Transit Duration (day)	0.095 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.149 ± 0.043 vs 0.1254 (deviation 0.55σ)
Duration fit vs published	0.095 d vs 0.0746 d (deviation 0.76σ)
Mid-time O–C	-1.4 min
Depth SNR	3.5
Residual scatter	8.3 %

Light curve

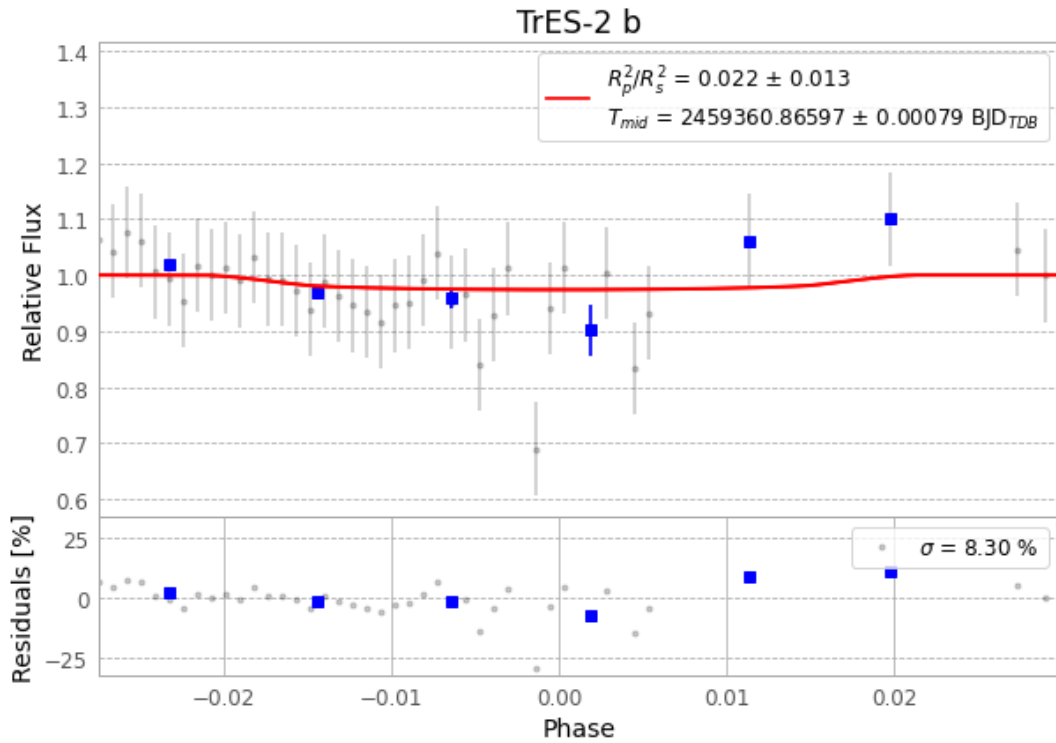


Figure 1 — FinalLightCurve_TrES-2 b_2021-05-26.png (SHA-256 469890adf3ab6d4c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [357, 239], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 914f535039d102dd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

70 FITS frames (46 MB), TRES-2210526070612.FITS → TRES-2210526103312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-210526.log (775bae7ac2aa...), FinalLightCurve_TrES-2 b_2021-05-26.png (469890adf3ab...), FinalLightCurve_TrES-2 b_2021-05-26.csv (dcaf736edcc4...)

Compute resources

Wall time 320.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 00:02Z.